Learning Discrete Representations from Mass Spectrometry Imaging using VQ-VAE

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May 18, 2025



A proposal submitted to the Faculty of Science, University of the Witwatersrand, Johannesburg, in partial fulfilment of the requirements for the degree of Master of Science

Abstract

Mass Spectrometry Imaging (MSI) has gained popularity in recent years as a result of the enhancements in technology that allow for capturing high-resolution mass spectra over spatial coordinates. Due to these enhancements, the data captured by MSI tools suffer the curse of dimensionality, making the data difficult to analyse and extract value from. Owing to this, efficient and effective data compression and dimensionality reduction techniques are necessary. The use of the VQ-VAE model is proposed as a possible solution to these needs, and the discrete latent space produced by the model is hypothesised to capture meaningful features at a lower dimensionality. These features could lead to an explainable model that help domain experts understand the most important factors of variation in MSI data.

Declaration

I, Jason Mark Wille, hereby declare the contents of this research proposal to be my own work. This proposal is submitted for the degree of Master of Science in Computer Science at the University of the Witwatersrand. This work has not been submitted to any other university, or for any other degree.

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Chapter 1

Introduction

Mass spectrometry imaging (MSI) is a technique that allows high-quality imaging of biological samples [Hu and Laskin 2022]. In recent years, MSI has seen many improvements and, according to Hu and Laskin [2022], is an important technique in biological research, clinical studies, drug discovery, and other fields. One of the advances in MSI has been an increase in spatial resolution, which has led to the need for better tools for data visualisation and data analysis [Zhang and Liu 2018].

According to Hu and Laskin [2022], the number of publications related to AI and MSI had increased exponentially over the period between 2016 and 2021. This is likely due to advances in computational efficiency, improvements in deep learning techniques, and a growing interest in their applications to different fields.

In MSI, mass spectra are collected that contain thousands of mass-to-charge (m/z) readings for each pixel of the image [Bjärterot *et al.* 2025]. This leads to high-dimensional data and the need for effective dimensionality reduction techniques that do not lose meaningful data, but that do remove unnecessary noise. These representation learning methods can be referred to as dimensionality reduction or data compression techniques and allow a more compact representation of the original data to be used to perform data analysis. This is desirable as it helps to extract the most useful signals from the data, which in turn allows for more accurate data analysis.

In order to obtain insights from MSI data, extensive data-preprocessing is needed and often the task of data analysis is broken up into the following three phases Hu and Laskin [2022]:

1. Spectral Data Processing
2. Data Compression
3. Data Interpretation

This research will primarily focus on the data compression phase. When compressing MSI data, it can be compressed in a spatial sense or in a spectral sense, where spatial compression allows for an overview of the spatial patterns, and spectral compression reveals the heterogeneity of biological samples. Self-supervised learning has shown

to be beneficial for this task as it does not require access to labelled data, which is challenging to obtain for spatial data. Deep learning has been shown to have the ability to learn both spectral and spatial components as well as their non-linear relationship [Manifold *et al.* 2021].

The use of machine learning algorithms for the analysis of MSI data has attracted interest in recent years in order to match the demand placed on techniques due to the massive amounts of data produced by an MSI experiment [Zhang and Liu 2018]. Van Den Oord *et al.* [2017] propose a machine learning model known as VQ-VAE, a self-supervised model that learns a discrete latent feature space that conserves the important features of the data. This discrete latent feature space represents a compressed form of the data. The VQ-VAE is a specialised version of the more general autoencoder. The standard autoencoder was first proposed for dimensionality reduction of MSI data by Thomas *et al.* [2016].

In this research, the specialised autoencoder model, VQ-VAE, will be used to learn discrete representations of MSI data in the hope that the representations will more accurately capture the factors of variation of MSI data. When reading the relevant literature, no use of applying the VQ-VAE model to MSI data was found.

The research proposal will start by introducing the necessary background and literature in Chapter 2. The background is split into Section 2.1 relating to deep learning and Section 2.2 relating to computational methods used in MSI. This will outline the current state of the field and set the scene for why the use of VQ-VAE could be relevant in the field of MSI.

Chapter 3 will formalise the research hypothesis, the purpose of the research and the research objectives. It will also mention how to determine the usefulness of the latent features obtained. Finally, Chapter 4 will outline a high level plan for the research in terms of a timeline to be followed, risk factors involved, and limitations of the research which may allow for future work.

Chapter 2

Background and Related Work

2.1 Deep Learning

The strength that deep learning (DL) provides is based on the deep artificial neural networks (ANNs) that are the foundation of most DL techniques. The term deep neural networks refers to an ANN with many hidden layers. The complexity added to the model by adding more hidden layers allows for complex non-linear relationships to be found in the data on which the model is being trained. LeCun *et al.* [2015] state that DL methods are representation learning methods that learn different representations built up from the raw input. Each representation adds extra complexity and allows for complex functions to be learnt from the data. Artificial neural networks will be discussed in Section 2.1.1.

One advantage of DL techniques when compared to conventional ML techniques is that DL techniques do not require handcrafted features to be used as input to their models. This means that complex features in the data can automatically be learnt by DL techniques. This saves a lot of time and effort; in the past domain expertise would be required in order to design useful features from complex datasets.

When training a deep learning algorithm there will be datasets that contain labels and others that do not. This leads to two different categories of ML techniques known as supervised and unsupervised learning. Supervised learning involves training a model in order to optimise some metric based on the labels provided for each data point. Unsupervised learning involves trying to learn something from the data itself without any labels.

One type of unsupervised learning technique, known as autoencoders, are of particular interest as they take data from a higher dimensional space and encode the data into a latent space that is usually a lower dimensionality than the original data. Autoencoders will be discussed in Section 2.1.2.

2.1.1 Artificial Neural Networks

Artificial neural networks are the foundational building blocks for deep learning techniques and are based on a perceptron which, in turn, is based loosely on a biological neuron. The nodes in a neural network take in inputs, multiply them by weights on the different edges and add a bias value. Once this is calculated, an activation function is applied and the output is optionally fed into the next layer.

Equation 2.1 shows how to calculate the output (also known as the activation) represented by a. The node at which the output is calculated is represented by j. A layer in the neural network is represented by l. A weight connecting node i in layer $l-1$ to node j in layer l is represented by $w_{ij}^{[l]}$. The bias at node j in layer l is represented by $b_j^{[l]}$. The activation at node i in layer $l-1$ is represented by $a_i^{[l-1]}$. Figure 2.1 shows the general structure of an artificial neural network and Figure 2.2 shows a visualisation of the calculation of an activation value.

$$a_j^{[l]} = g_j^{[l]}\left(\left(\sum_i a_i^{[l-1]} w_{ij}^{[l]}\right) + b_j^{[l]}\right) \tag{2.1}$$

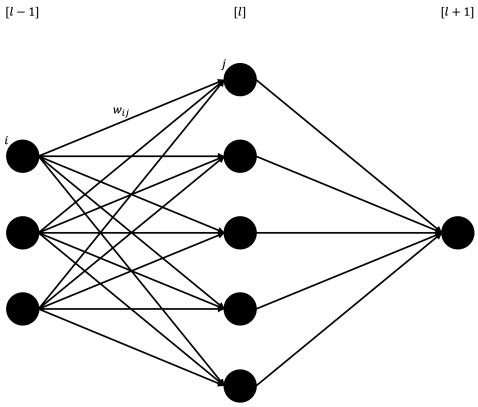


Figure 2.1: Structure of a fully-connected ANN with 3 layers

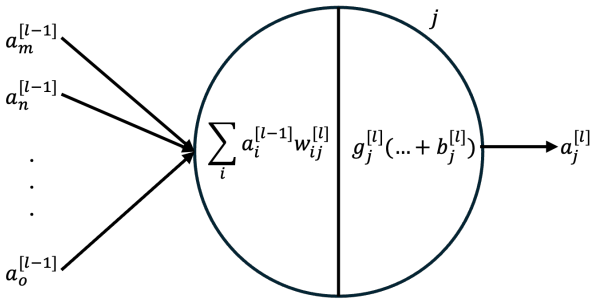


Figure 2.2: Activation calculation for node j in layer l with i input activations from layer $l-1$

Inputs are fed into an ANN and the calculation shown in Equation 2.1 is done for the nodes starting from the first layer and continuing until the final layer. This results in an

output for the whole ANN. This output, as well as the label associated with the input, can be used to calculate a loss. This loss is then used to calculate gradients for all of the weights and bias values using the partial derivatives of the loss with respect to each parameter. The gradients are used to slowly adjust the parameters and this process is repeated until the loss has reached an acceptable value.

2.1.2 Autoencoders

Autoencoders are a specific class of ANN which do not relate an input to a target but rather attempt to take an input, compress it into a lower dimensional space and then reproject the lower dimensional value to the original input. Autoencoders are an unsupervised technique, meaning that they do not require labelled data in order to learn. Hinton and Salakhutdinov [2006] states that the autoencoder is a non-linear generalisation of principal component analysis (PCA). Autoencoders are made up of two components, an encoder, and a decoder. The encoder compresses the data to a dimensionality that can fit through the bottleneck and the decoder takes the data from the bottleneck and tries to reproduce the input fed into the encoder. The error is measured as a reconstruction loss between the reconstructed data that the decoder produced and the input data. The architecture of a general autoencoder is shown in Figure 2.3.

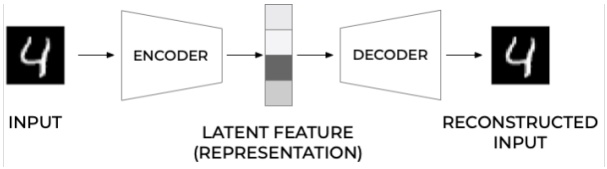


Figure 2.3: Architecture of an general autoencoder [Michelucci 2022]

The latent space of an autoencoder can be interesting to analyse as it can highlight aspects of the original data that form the most important features. One downside of the latent space of a regular autoencoder is that there are no constraints placed on it. One possible use of an autoencoder is to sample points from the latent space and pass them through the decoder in order to generate data points. This task is not feasible with a standard autoencoder but one model that does have this generative property is a variational autoencoder which will be introduced in Section 2.1.3.

2.1.3 Variational Autoencoder

The variational autoencoder (VAE) was introduced by Kingma *et al.* [2013] and the general structure of a VAE is shown in Figure 2.4. The VAE is similar to a standard autoencoder in the sense that it also has an encoder and a decoder portion (represented as "E" and "D" respectively in Figure 2.4), but different in the sense that the encoder is used to obtain a different μ and σ for each data point which represent the parameters for a Gaussian distribution.

Due to the randomness introduced by sampling from a distribution in the bottleneck, the loss function would not have been differentiable if it was not for Kingma *et al.*

[2013] introducing the reparameterization trick. The reparameterization trick is highlighted in Figure 2.4 which shows how the variability is obtained from a term, ϵ. This trick again allows the loss function to be differentiable and for backpropagation to be applied.

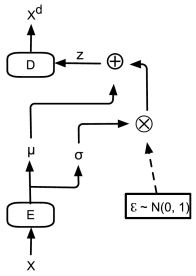


Figure 2.4: Structure diagram of a VAE [Li *et al.* 2023]

When training a VAE we are interested in learning different distributions. The data comes from a distribution $p(x)$, while the distribution of the latent variable z comes from $p(z)$. We would like to also know the distribution $p(z|x)$ as it would allow us to get the latent variable z for a given data point x. These conditional and marginal distributions are linked using Bayes' theorem shown in Equation 2.2 and the glossary for the names of these distributions is shown in Table 2.1.

$$p(z|x) = \frac{p(x|z)p(z)}{p(x)} \tag{2.2}$$

Table 2.1: The different parts of Bayes' theorem [Odaibo 2019]

Symbol	Name
z	Latent variable
$p(z)$	Prior probability
$p(z\|x)$	Posterior probability
x	Evidence or data
$p(x)$	Evidence probability
$p(x\|z)$	Likelihood probability

When training a VAE we obtain an approximation of the posterior distribution, which is denoted as $q_\theta(z|x)$. θ represents the distribution parameters and can be thought of as the weights of the encoder portion of the network. The decoder portion of the network and its weights represent the likelihood distribution denoted as $p_\phi(x|z)$. This is shown in Figure 2.5.

We want the approximation of the posterior distribution, $q_\theta(z|x)$, to be as close to the posterior distribution, $p(z|x)$, as possible. The KL divergence is used for this and is shown in Equation 2.3.

$$D_{KL}[q_\theta(z|x)||p(z|x)] = E_{q_\theta(z|x)}[\log(q_\theta(z|x)) - \log(p(z|x))] \tag{2.3}$$

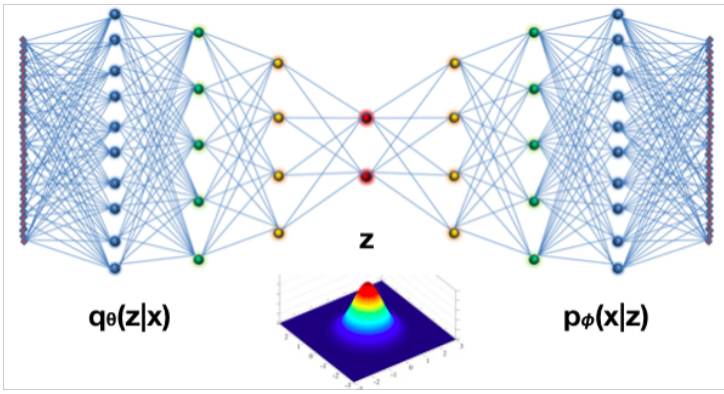


Figure 2.5: Distributions learnt by the encoder and decoder of a VAE [Odaibo 2019]

Equation 2.3 can be simplified to show that the cost for the VAE can be represented as Equation 2.4. This derivation is shown in an understandable manner by Odaibo [2019].

$$J_{\text{VAE}} = -D_{KL}[q_\theta(z|x)||p(z)] + E_{q_\theta(z|x)}[\log p_\phi(x|z)] \tag{2.4}$$

The first term of the loss function is used to constrain the shape of the posterior approximation. The second term of the loss function is called the reconstruction term [Odaibo 2019] because it measures the likelihood of the data reconstructed by the decoder.

2.1.4 VQ-VAE

The VQ-VAE was introduced by Van Den Oord *et al.* [2017] and differs from a VAE in two ways:

1. The encoder network outputs discrete codes, instead of continuous ones.
2. The prior $p(z)$ is learnt rather than fixed.

In the case of VAEs, often the posterior and prior distribution were assumed to be normally distributed. In the case of the VQ-VAE, where discrete latent variables are used, the posterior and prior distribution are categorical and samples from these distributions index an embedding table, also known as the codebook [Van Den Oord *et al.* 2017].

In general, a VQ-VAE has three trainable parameters. The encoder, the decoder, and the codebook. This codebook constraints the latent space to be discrete and is a variation of a standard VAE.

Discrete Latent Variables

The latent embedding space used in a VQ-VAE is defined as a space $e \in R^{K \times D}$. This means that there will be K embedding vectors $e^i \in R^D$ where $i \in 1, 2, ..., K$. The model takes in an input x, passes it through the encoder which outputs $z_e(x)$. These values

then use the codebook to find the nearest embedding vectors which will be used as input to the decoder. This step is shown in Equation 2.5.

$$z_q(x) = e_k, \text{ where } k = \text{argmin}_j ||z_e(x) - e_j||_2 \tag{2.5}$$

In order to work the above process into the VAE framework, the posterior distribution can be represented as in Equation 2.6. This uniform prior distribution means that the KL divergence term is a constant and equal to $\log K$. It follows that the KL divergence term is not needed in the loss function of the VQ-VAE.

$$q(z = k|x) = \begin{cases} 1 \text{ for } k = \text{argmin}_j ||z_e(x) - e_j||_2 \\ 0 \text{ otherwise} \end{cases} \tag{2.6}$$

Learning

The loss function for the VQ-VAE is shown in Equation 2.7. The loss function can be thought of as comprising of three terms.

$$L = \log p(x|z_q(x)) + ||\text{sg}[z_e(x)] - e||_2^2 + \beta||z_e(x) - \text{sg}[e]||_2^2 \tag{2.7}$$

The first term of the loss function is the reconstruction term which is used to tune the weights of the encoder and the decoder. Due to the fact that the quantization process is not differentiable, the gradients are passed straight through from $z_q(x)$ to $z_e(x)$, this means that there are no weights to update the codebook.

The second term of the loss function uses Vector Quantisation (VQ) to move the embedding vector e_i towards the output of the encoder $z_e(x)$. The sg function represents the stop-gradient function which highlights that the encoder should not be updated with these weights and only the embedding vector should be updated.

The third term of the loss function is known as the commitment term and penalises the encoder if the encodings stray too far from their codebook embeddings.

The different components of the VQ-VAE are show in Figure 2.6.

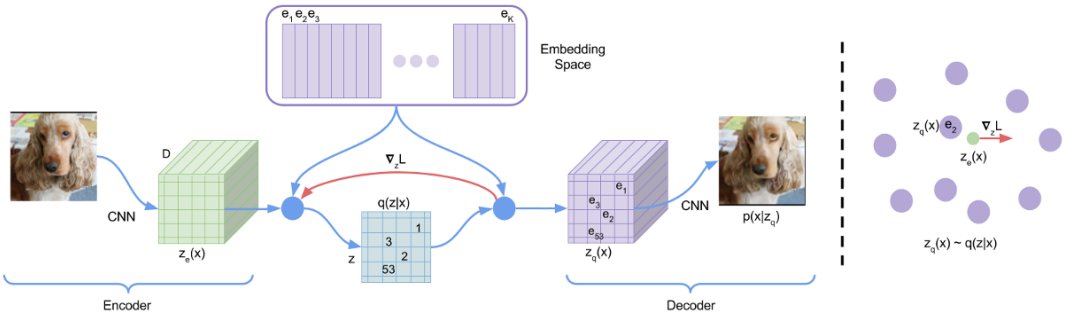


Figure 2.6: Structure of a VQ-VAE [Van Den Oord *et al.* 2017]

2.2 Computational Methods in MSI

2.2.1 Mass Spectrometry Imaging

Mass spectrometry imaging (MSI) is a technique that allows high-equality imaging of biological samples [Hu and Laskin 2022]. MSI data is captured by taking mass spectra in different x, y coordinates in order to form a hyperspectral image. The general process followed in an MSI experiment is shown in Figure 2.7. An experiment starts by collecting the sample that will be used and then preparing it for the MSI instrumentation that will be used to capture the spectra. The next phase of the experiment involves capturing the spectra of the sample at different x, y coordinates and then processing and visualising the raw data. Finally, the data processing section of the experiment is the last phase where different computational techniques can be applied to the data collected.

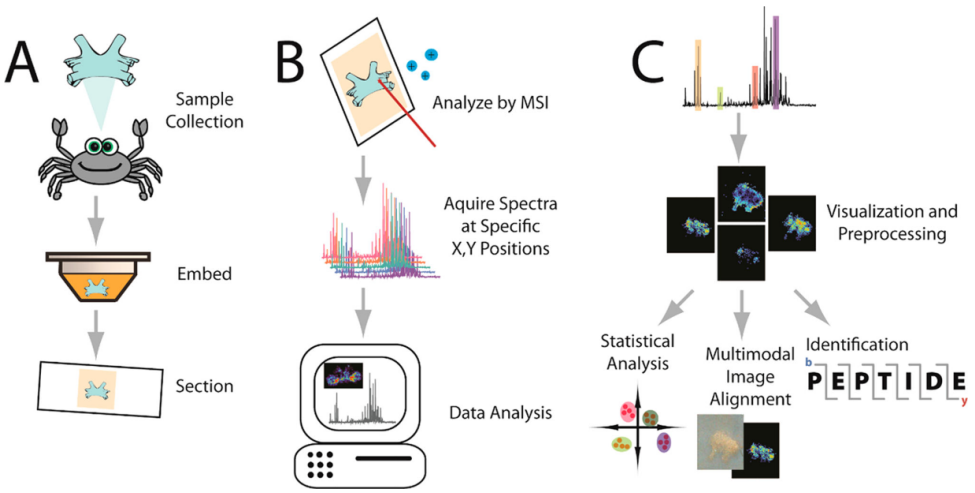


Figure 2.7: Procedure for a general MSI experiment [Buchberger *et al.* 2017]

According to Fonville *et al.* [2013], most biochemical analyses do not take into account spatial distributions, which is vital for biomedical research. This is where the strength of MSI comes into play, as MSI can represent both spatial and spectral information by measuring different mass-to-charge spectra over a spatial grid. MSI data can be interpreted as mass spectra at a specific x, y coordinate (or pixel), or it can be interpreted as an image of a specific ion's intensity over a 2D set of pixels; this is highlighted by the general structure of the MSI data cube shown in Figure 2.8.

In recent years there have been advances in the spatial resolution and acquisition throughput of MSI [Hu and Laskin 2022], this has led to an increase in the size of a single data point captured in an MSI experiment. This increase in the size of the data has highlighted the need for more efficient methods for data visualisation and analysis.

2.2.2 Data Compression

The 3D data cube shown in Figure 2.8 is often considered for data compression after spectral data processing has been carried out on the data. The spectral data processing

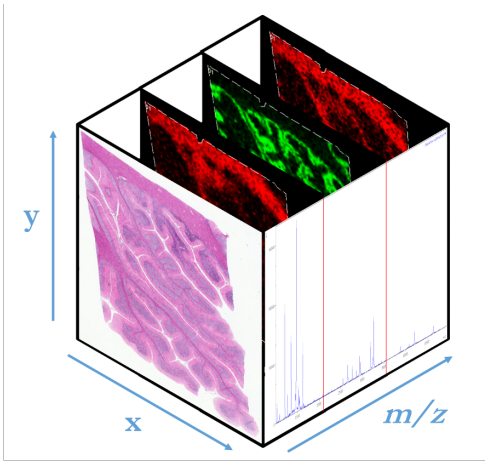


Figure 2.8: Data cube highlighting the spatial and spectral features of MSI data [Galli and others 2018]

tries to remove some noise from the data by using methods such as peak picking and signal normalisation. Even after this noise has been removed from the data, the data still has high dimensionality. This leads to the need for useful data compression and dimensionality reduction techniques.

According to Hu and Laskin [2022], matrix factorisation techniques such as PCA, non-negative matrix factorisation and multivariate curve resolution-alternating least squares have been used extensively. Methods such as self-organised maps, t-distributed stochastic neighbour embedding (t-SNE), and uniform manifold approximation and project (UMAP) have also gained popularity. Self-supervised learning has also gained popularity in the space of deep learning models as it avoids the need to have labelled data. Thomas *et al.* [2016] were the first to propose an autoencoder architecture to compress spectral data which was followed by Abdelmoula *et al.* [2021] who extended their strategy and managed to avoid the need for spectral data processing by making use of a VAE.

One of the benefits found when using neural network based approaches compared to t-SNE and UMAP is that neural network based approaches allow for models to be trained in batches which allows for only portions of the large MSI data to need to be present in RAM at a time [Michelucci 2022].

Another consideration is that data can be compressed in a spectral and a spatial manner. There has been more research into methods of compression for spectral data, but spatial compression has not been well explored. Hu and Laskin [2022] predict that self-supervised learning will provide enhanced computational methods for MSI data in the future.

Chapter 3

Methods

3.1 Hypothesis

The use of autoencoders for the compression of MSI data has proven to be useful for later downstream tasks but due to the disorganised latent spaces of regular autoencoders and the low interpretability of variational autoencoders, the use of a VQ-VAE for the compression of MSI data could lead to a highly interpretable and low dimensional latent space.

The ideal latent space would represent a few unique molecules in each spatial location. This would effectively reduce the spectral dimensionality and allow for clearer understanding of spatial patterns.

3.2 Research Questions

- Can discrete latent representations learnt by VQ-VAE from MSI data reduce the dimensionality of the data while still preserving the most important features?
- Are the latent representations in any way disentangled? If so, do they lead to an explainable representation of the original data?
- To what extent can the spectral component of MSI data be compressed while still learning useful representations?

3.3 Objectives

The objective of this paper will be to highlight that the use of a VQ-VAE leads to a useful latent representation of MSI data. Determining if a latent space is useful can be a difficult task and there are different metrics that can be used to quantify their usefulness. Another consideration is that the usefulness of a latent space can depend on what the latent representations will be used for [Baykal *et al.* 2024]. Due to this reason it is useful to have a benchmark to compare results against. Unfortunately,

the use of VQ-VAE's have not been explored for MSI data, however, Abdelmoula *et al.* [2021] proposed a model called msiPL which is based on a fully connected VAE. Their model was used for dimensionality reduction of MSI data and will be used as a benchmark when comparing results.

The idea of disentangled representation learning is another useful concept when learning representations of data. Disentangled representations lead to more explainable models [Wang *et al.* 2024]. The idea is that the latent space of the VQ-VAE should represent different factors of variation. These factors of variation should represent underlying factors that are hidden in the data. Figure 3.1 highlights the usefulness of knowing the different factors of variation. Knowing them could allow for the activation of certain independent factors and observations of the differences in the output of the model.

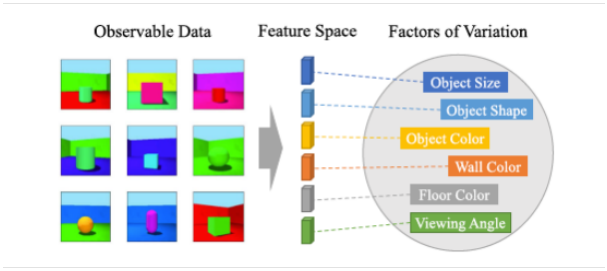


Figure 3.1: The usefulness of learning factors of variation [Wang *et al.* 2024]

In order to understand the performance of the VQ-VAE model on learning a disentangled representation, the DCI disentanglement metric will be used [Eastwood and Williams 2018].

Another way that the models will be compared is by performing latent traversals where different codes from the VQ-VAE's codebook will be used as input to the decoder in order to see the decoded output. Once these decoded outputs are produced, they can be examined by the domain experts to understand whether or not the codes are producing meaningful differences in the output. The decoded output can also be compared to the output from the VAE proposed by Abdelmoula *et al.* [2021] to determine if the VQ-VAE is making a meaninful contribution in comparison to a VAE.

The learning speed of the VQ-VAE can also be compared to that of the VAE. While knowing whether the representations that the different models learnt are useful or not, it will also be interesting to note whether the VQ-VAE or VAE take longer to train. It will be beneficial for a model to train faster as the size of the data will make it prohibitive to train for many epochs.

The codes learnt by the VQ-VAE will also be compared with the compounds in the Lipid Maps and HMDB databases. If the codes are highly correlated to certain compounds in those databases it could prove useful to know what each compound represents in the original data.

Chapter 4

Research Plan

4.1 Timeline

The general plan for the research is to operate in an iterative manner that allows for improvements on results to be explored. The three general phases of spectral data processing, data compression and data interpretation will be iterated on in a three week cycle. This will allow for 5 iterations of the cycle by the time the deadline has arrived.

Unforeseen circumstances may only allow for four iterations, but this still qualifies as being acceptable. The first iteration will require no work on the final research report as this will happen when most of the experiment is set up.

Once this initial set up has been completed, it will be easier to iterate and write up about the potential improvements.

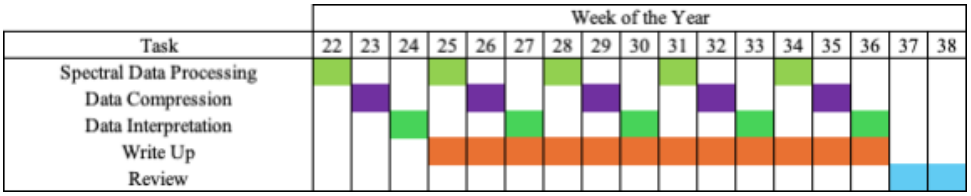


Figure 4.1: Research plan represented as a gant chart

4.2 Deliverables

The main deliverable for this research will be a research report that includes further information on the methods followed in order to conduct the experiments proposed in this proposal. Another deliverable will be the code used to run the experiments performed for this research, as well as the output and data analysis provided by the code.

4.3 Risk Factors

Inadequate Preprocessing

Spectral data processing is one of the steps in the data analysis of MSI data and is used to ensure that there is as little noise as possible present in the data. Due to the curse of dimensionality [Hu and Laskin 2022] it can be difficult to extract a useful signal from the noise in the data. Deep learning methods should be able to learn useful features at a hidden layer and therefore spectral processing will be avoided.

This could lead to useful results in the sense that not needing to perform spectral processing that requires domain expertise will allow for a more general framework that can be more easily applied. This could also result in a bottleneck in the performance of the VQ-VAE model. If the deep learning model is not able to learn useful features then spectral processing can be applied to examine the extent to which it can improve the performance of the VQ-VAE model.

4.4 Limitations

The research proposed aims to explore the latent space of MSI data created by a VQ-VAE model. This latent space may produce interesting properties due to the quantised nature of the codebook, however, the usefulness of the latent space would need to be measured against different downstream tasks such as classification in order to know if the latent space is well suited for data analysis.

Chapter 5

Conclusion

Mass spectrometry imaging provides a method that does not require prior tagging of molecular targets and can also measure multiple ions in a single experiment [Verbeeck *et al.* 2020]. For these reasons it has proven useful for exploratory analysis in a range of domains. Even though being able to measure multiple ions in a single experiment has proven useful, it presents a challenge in deriving useful information from the data collected in an MSI experiment. This is due to a phenomenon known as the curse of dimensionality [Prasad *et al.* 2022].

The curse of dimensionality can be solved by using effective dimensionality reduction techniques that map the original data from their input dimensionality to a more compact dimensionality. This lower dimensionality space needs to accurately capture the essence of the original data. In recent years, different methods have been applied to the task of dimensionality reduction on both the spectral and spatial aspects of MSI data.

Thomas *et al.* [2016] was the first to propose the use of standard autoencoders for dimensionality reduction of MSI data. Abdelmoula *et al.* [2021] developed a framework based on a variational autoencoder that circumvented the need for spectral data processing. Thomas *et al.* [2016] mentions that most dimensionality reduction techniques applied are linear and do not have the ability to learn non-linear mappings. This is overcome by the use of autoencoders. The use of autoencoders also allowed for less human intervention in the analysis of MSI data.

Kingma *et al.* [2013] introduced the variational autoencoder which allowed for autoencoders to be seen as a generative model and Van Den Oord *et al.* [2017] introduced a variation of this model known as the VQ-VAE. The VQ-VAE allows for a discrete latent distribution of the autoencoder to be learnt whilst still maintaining the generative abilities of the model. This discrete latent space, also known as the codebook, could provide insights into the structure of MSI data and will be explored in this research.

Chapter 6

AI Statement

I understand that the use of generative AI tools (such as ChatGPT or similar) without explicitly declaring such use constitutes a form of plagiarism and is classified by Wits University as academic misconduct.

I declare that in the course of conducting the research towards my degree or in the preparation of this research proposal **I did make use of generative AI tools for the following**:

Generative AI Tool Used	Usage
ChatGPT	Simplifying complex concepts from research papers.
ChatGPT	Generating LaTeX code.
Google Scholar PDF Reader	AI outline of research papers.

If generative AI tools were used as an integral part of the experimental design or in the direct execution of my research, I confirm that details of this use are clearly outlined in the relevant experimental/methodology chapters of my research proposal.

Jason Mark Wille
University of the Witwatersrand

Date: May 18, 2025

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